EXPERIMENT NO: 09

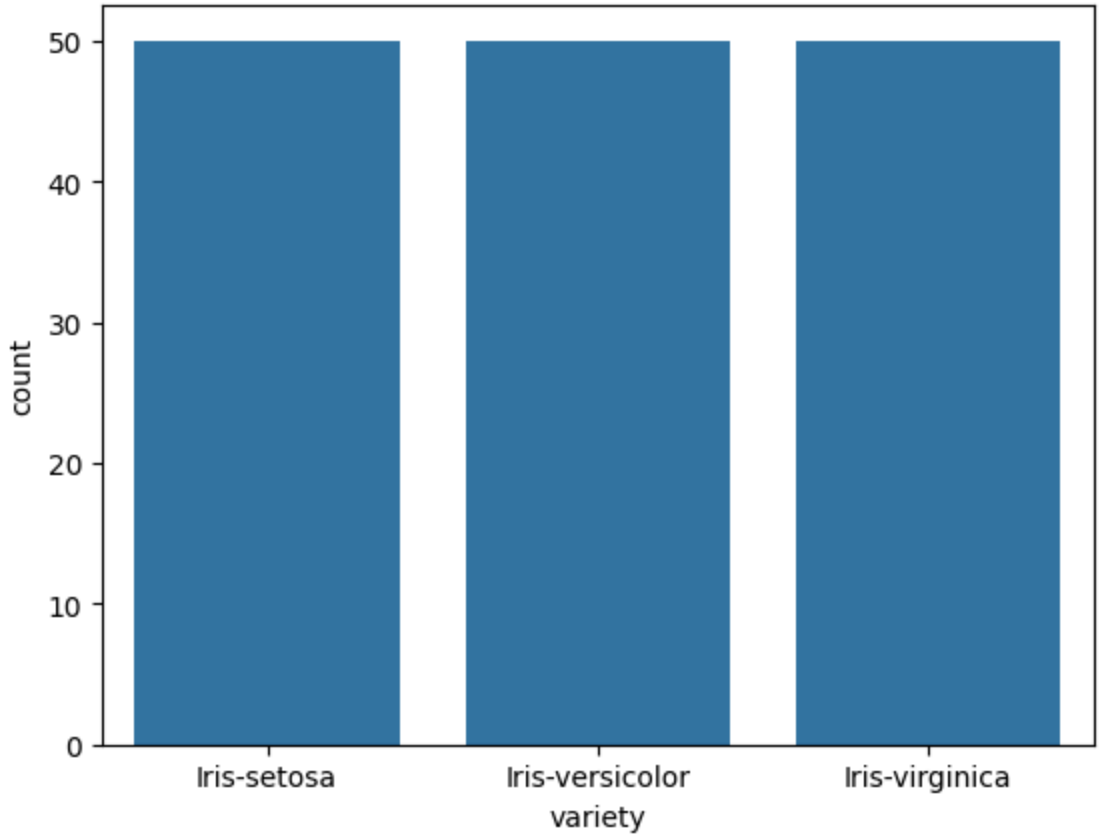
Experiment to understand Matplotlib library use cases in Data Science through visualization

Program:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv('/content/Iris_Dataset.csv')
data.info()
print(data.describe())
print(data.value_counts('variety'))
sns.countplot(x='variety',data=data,)
plt.show()
dummies=pd.get_dummies(data.variety)
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]
]],axis=1)
print(FinalDataset.head())
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data
,)
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data
,)
sns.pairplot(data,hue='variety',height=3);
plt.show()
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm
').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm'
).add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm
').add_legend();
plt.show();
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm'
).add_legend();
plt.show();
```

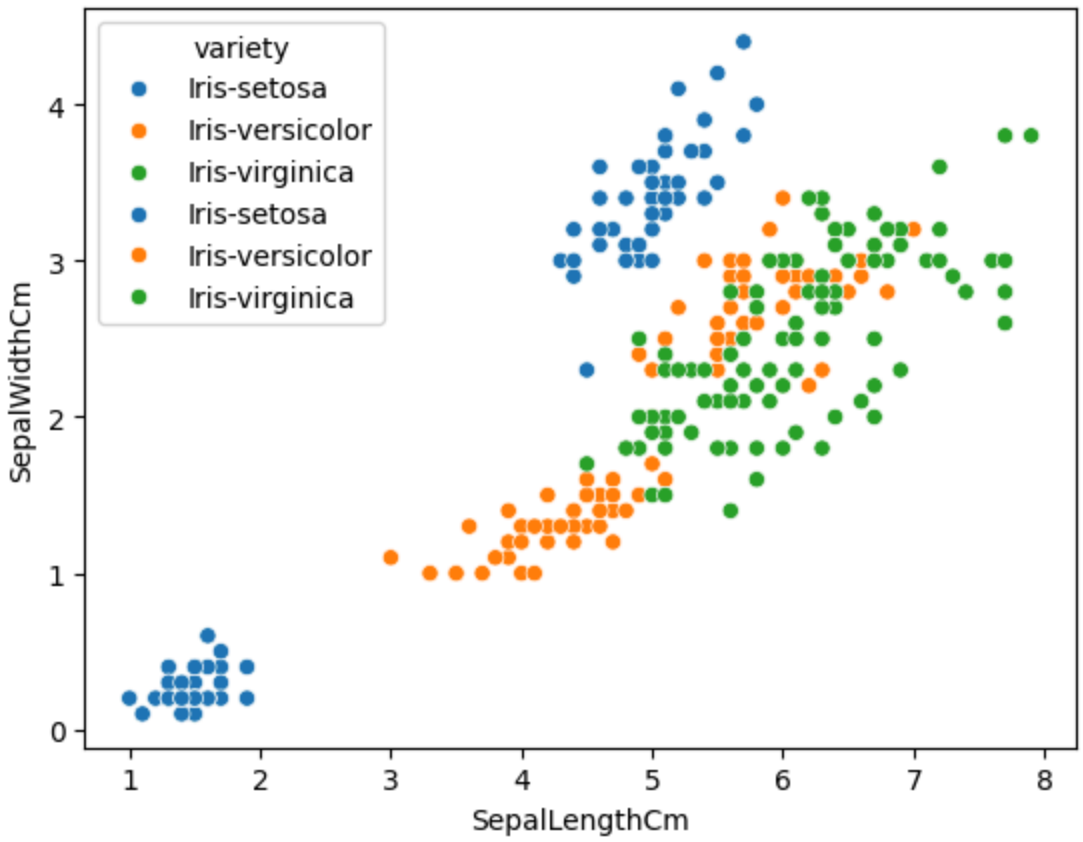
Output:

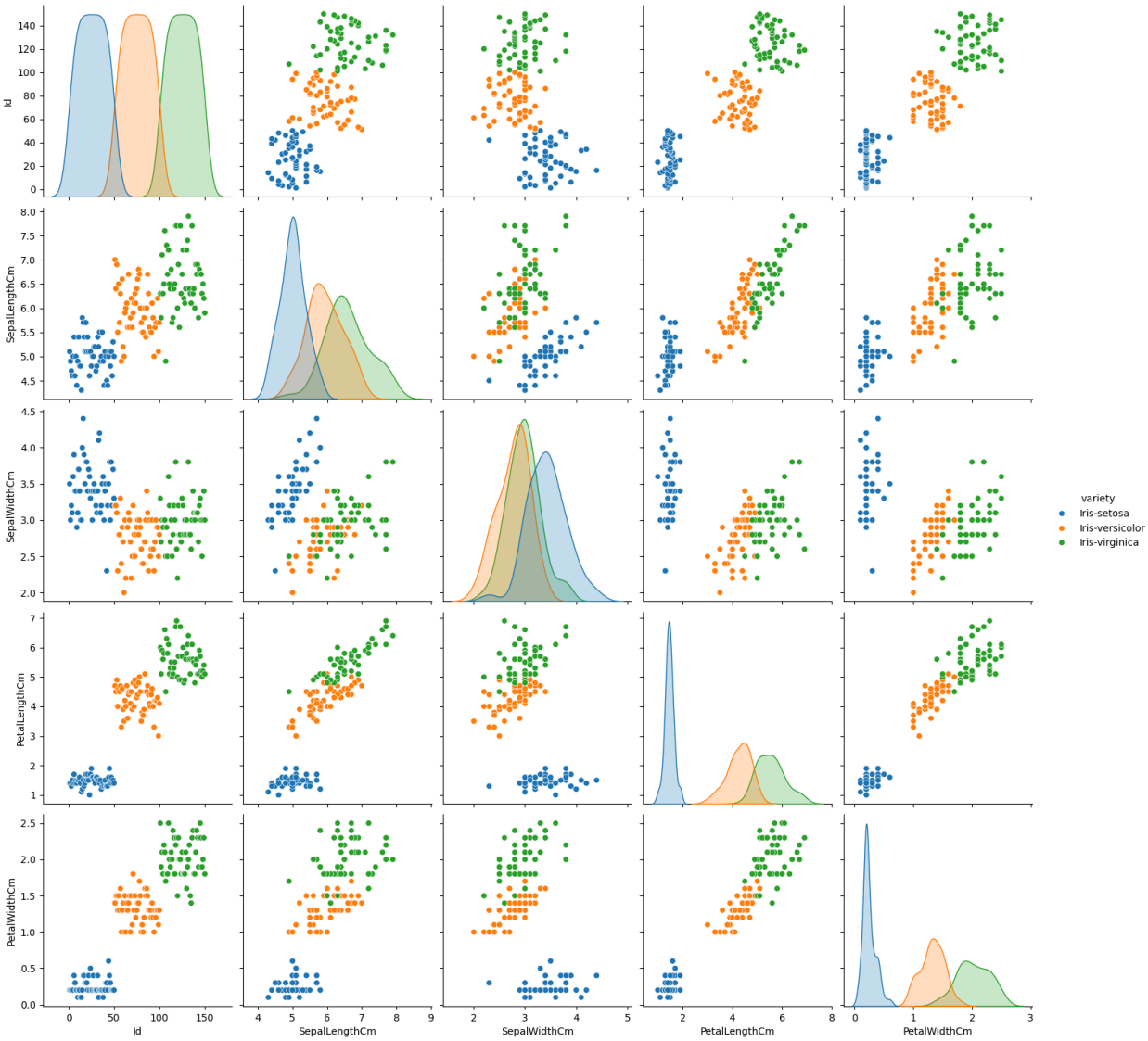
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
               Id  SepalLengthCm  SepalWidthCm  PetalLengthCm
PetalWidthCm
count  150.000000     150.000000    150.000000     150.000000
150.000000
mean    75.500000       5.843333      3.057333       3.758000      1.199333
std     43.445368       0.828066      0.435866       1.765298      0.762238
min      1.000000       4.300000      2.000000       1.000000      0.100000
25%     38.250000       5.100000      2.800000       1.600000      0.300000
50%     75.500000       5.800000      3.000000       4.350000      1.300000
75%    112.750000       6.400000      3.300000       5.100000      1.800000
max    150.000000       7.900000      4.400000       6.900000      2.500000
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

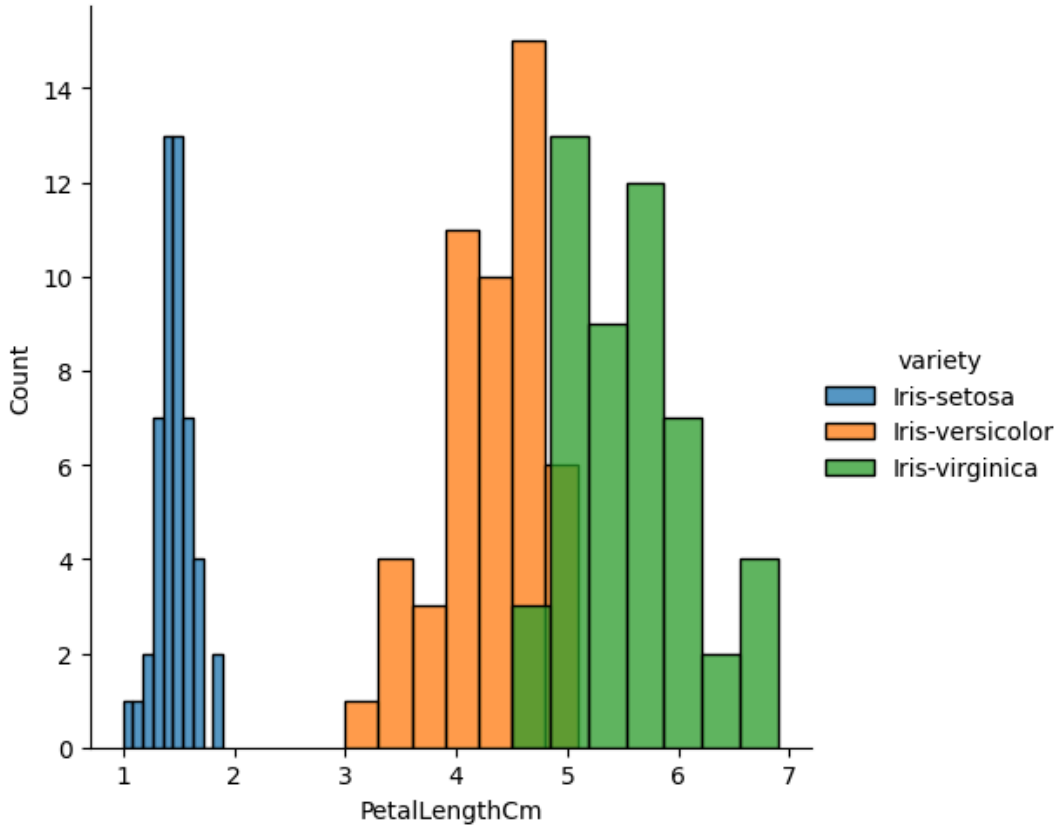


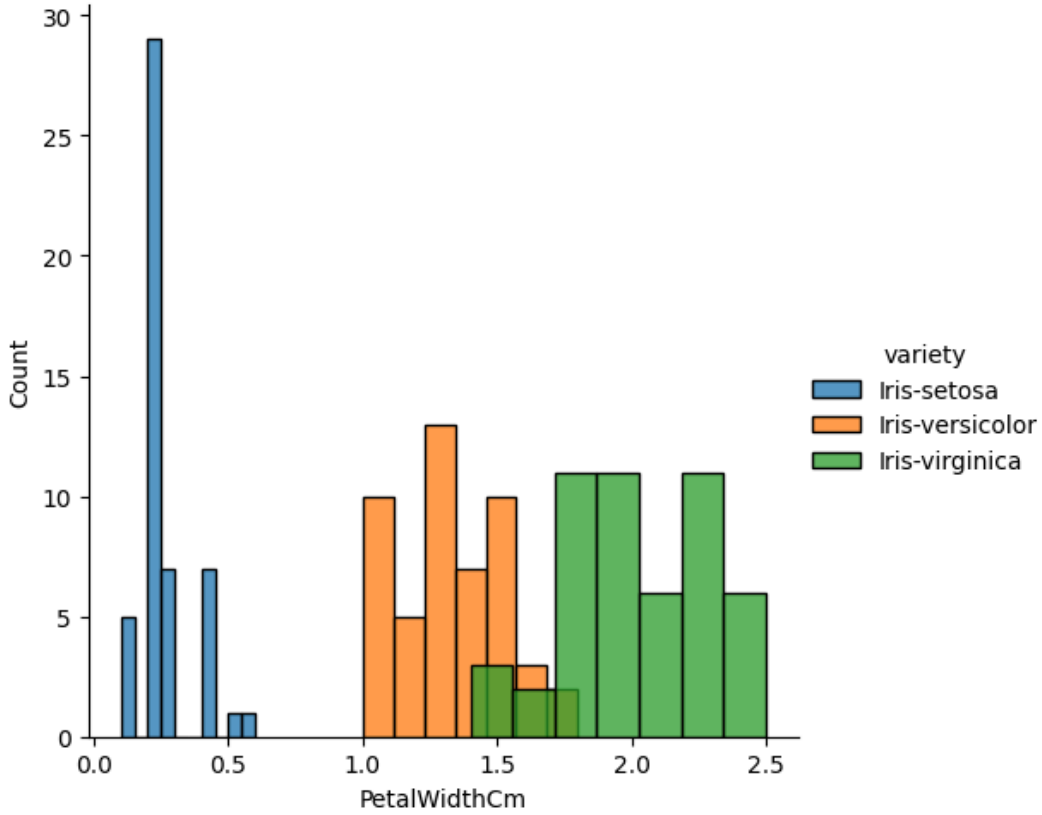
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	\
0	True	False	False	1	5.1	
1	True	False	False	2	4.9	
2	True	False	False	3	4.7	
3	True	False	False	4	4.6	
4	True	False	False	5	5.0	

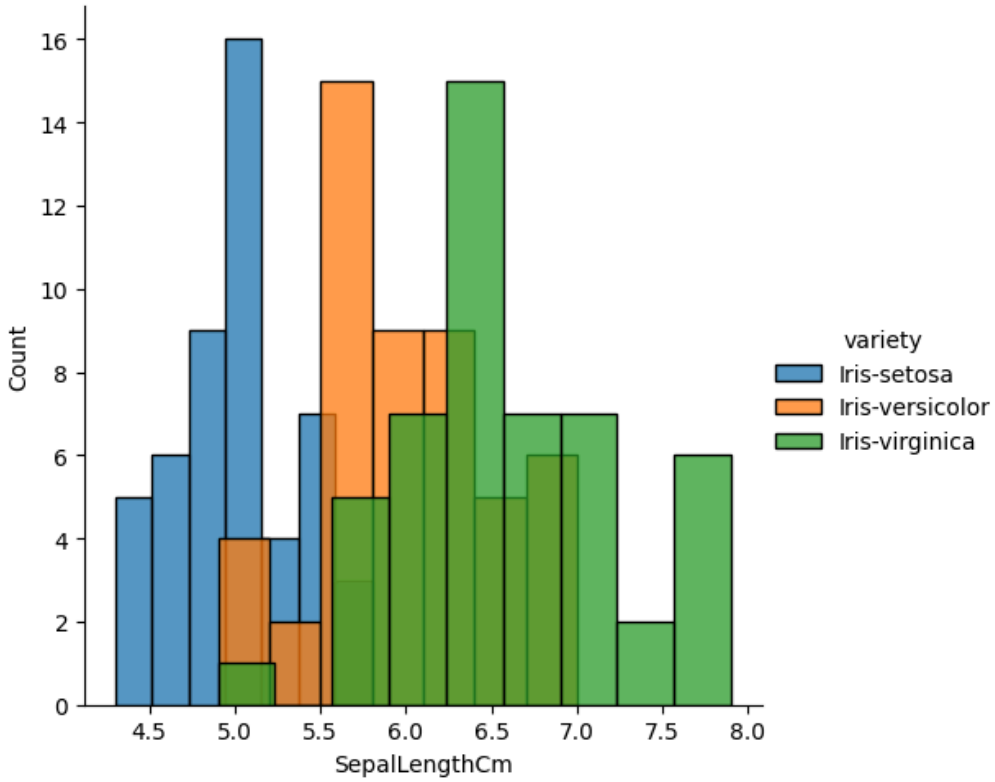
	SepalWidthCm	PetalLengthCm
0	3.5	1.4
1	3.0	1.4
2	3.2	1.3
3	3.1	1.5
4	3.6	1.4

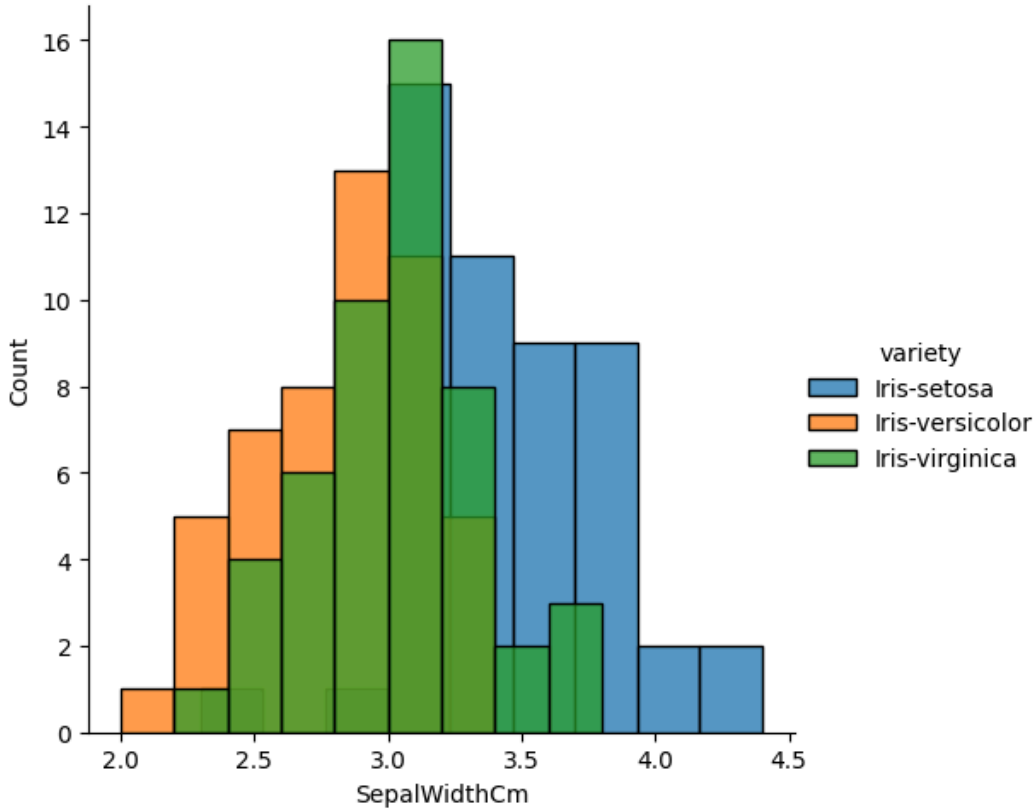

variety
Iris-setosa
Iris-versicolor
Iris-virginica
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
0
1
2
3
4
1
2
3
4
5
6
7
8


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica